16/10/25

Normal Distribution

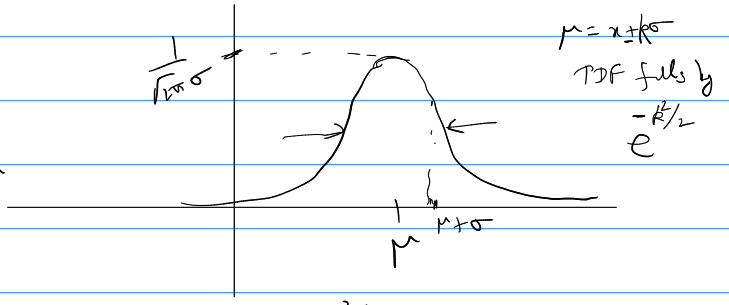
$X \sim N(\mu, \sigma^2)$ — μ: mean, $\mu \in \mathbb{R}$; σ^2: variance, $\sigma^2 > 0$ real; σ: standard deviation

PDF $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $x \in \mathbb{R}$

$E[X] = \mu$

$Var(X) = \sigma^2$

$E[X^2] = \mu^2 + \sigma^2$



$Z = \frac{X-\mu}{\sigma}$ PDF $f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, $z \in \mathbb{R}$

mean = 0, std dev = 1, variance = 1

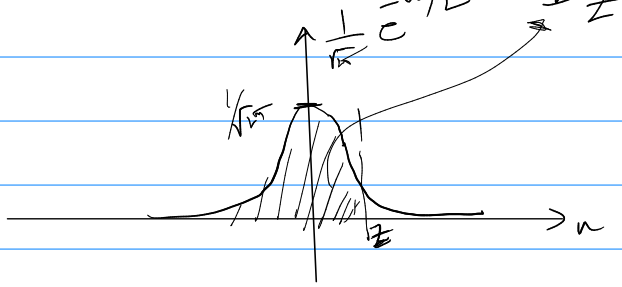
Standard Normal

" $N(\mu,\sigma) = \sigma N(0,1) + \mu$

$$\int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}} dx = \mu$$

$$\int_{-\infty}^{\infty} x^2 \cdot \; dx = \mu^2 + \sigma^2$$

CDF of $N(0,1)$ $\quad \Phi_Z(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$



Φ_Z^{-1} : assume known

$P(Z \le z) = \Phi_Z(z)$

$P(Z > z) = 1 - \Phi_Z(z)$

$P(Z \le -z) = \Phi_Z(-z)$

$= 1 - \Phi_Z(z)$

$1 - \Phi(1.96) \approx 0.025$

$X \sim N(\mu, \sigma^2)$

$$\Pr(X > x_0) = 1 - \Phi_Z\left(\frac{x_0 - \mu}{\sigma}\right)$$

$$\Pr\left(\mu - \sigma(1.96) < X < \sigma\cdot(1.96) + \mu\right) \approx 0.95$$

$$\Pr\left(\mu - \sigma\cdot\Phi_Z^{-1}(1-\varepsilon/2) < X < \mu + \sigma\cdot\Phi_Z^{-1}(1-\varepsilon/2)\right) = 1 - \varepsilon$$

$X \sim N(\mu, \sigma^2)$ $\quad Z \sim N(0,1)$

$$M_X(t) = E\left[e^{(\sigma Z + \mu)t}\right] \qquad M_Z(t) = e^{t^2/2}, \ \forall t$$

$$= e^{\mu t} M_Z(t\sigma)$$

$$= e^{\mu t + \sigma^2 t^2/2}$$

$X_i \sim N(\mu_i, \sigma_i^2)$, $i = 1, \ldots, n$, indep

$$Y = X_1 + X_2 + \cdots + X_n$$

$$M_Y(t) = \prod_{i=1}^{n} e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} = e^{(\mu_1 + \cdots + \mu_n)t + \frac{(\sigma_1^2 + \cdots + \sigma_n^2)t^2}{2}}$$

$$Y \sim N(\mu_1 + \cdots + \mu_n, \ \sigma_1^2 + \cdots + \sigma_n^2)$$

$\mu_1 = \cdots = \mu_n = \mu$, $\sigma_1^2 = \cdots = \sigma_n^2 = \sigma^2$

Sample mean $\quad \bar{X}_n = \dfrac{X_1 + \cdots + X_n}{n} \sim N\left(\mu, \dfrac{\sigma^2}{n}\right)$

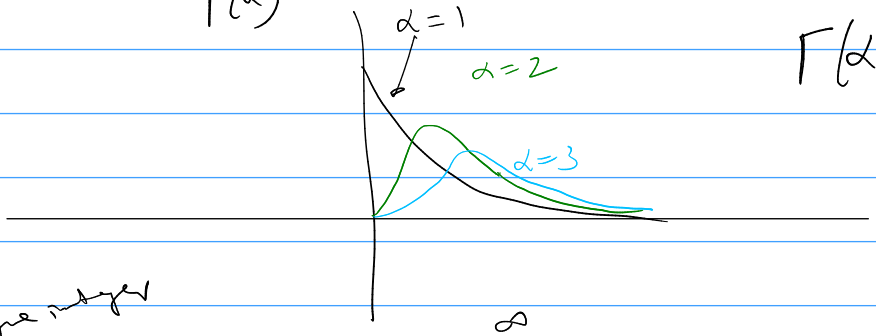
Gamma distribution

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx$$

$X \sim \text{Gamma}(\alpha, \beta)$ $\quad \alpha > 0$, $\beta > 0$

$$\frac{\Gamma(\alpha)}{\beta^\alpha} = \int_0^\infty x^{\alpha-1} e^{-\beta x}\, dx \qquad u = \beta x \qquad \beta > 0$$

PDF $f_X(x) = \dfrac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, $x > 0$



$\Gamma(\alpha) = (\alpha-1)\,\Gamma(\alpha-1)$

$\Gamma(1/2) = \sqrt{\pi}$

$\Gamma(n) = (n-1)!$

k: integer

$$E[X^k] = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{k+\alpha-1} e^{-\beta x}\, dx = \frac{\Gamma(k+\alpha)}{\Gamma(\alpha)\,\beta^k}$$

$$M_X(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty e^{tx} x^{\alpha-1} e^{-\beta x}\, dx = \left(\frac{\beta}{\beta - t}\right)^\alpha, \quad \beta > t$$

$X_i \sim \text{Gamma}(\alpha_i, \beta)$ $\quad i = 1, \dots, n$, indep.

$$Y = X_1 + X_2 + \cdots + X_n \sim \text{Gamma}(\alpha_1 + \cdots + \alpha_n, \beta)$$

(MGF)

$\alpha = 1$: Exponential (β) $\qquad$ PDF $\beta e^{-\beta x}$, $x > 0$

"Queuing theory"

Beta distribution

$$f_X(x) \propto x^{\alpha-1} (1-x)^{\beta-1}$$

$x \in [0, 1]$